We use the original $SU(N)$ Wilson measure, for fixed $N \geq 2$, on a four-dimensional periodic lattice with spacing $a$ and $n$ sites per direction:

$$d\mu_\beta = Z_\beta^{-1} \rho \, d\lambda, \qquad \rho = e^{-\beta S_0}, \qquad S_0 = \sum_{p \in \mathcal{P}} s_p, \qquad s_p = N - \Re \operatorname{tr} U_p, \qquad \beta = c_\beta g^{-2}. \tag{1}$$

Here $d\lambda$ is the product of normalized Haar measures and $c_\beta > 0$ is fixed. A coordinate $i = (b, A)$ includes its original oriented bond and Lie-algebra component. The invariant derivative $X_i$ is skew-adjoint with respect to Haar measure. All products of these derivatives remain in their displayed order.

The local moment statements below use precisely the reflection-compatible tori of [W], Chapter 10. Fix $K < \infty$ and impose

$$\beta \geq 2, \qquad \frac{\log \beta}{n} \leq K, \tag{2}$$

with $n$ large enough for the fixed contact stencils. This includes the usual fixed physical volume regime $n = T/a$ and $\beta = O(\log(1/a))$ along compatible lattice subsequences. We do not claim the same estimates uniformly as $\beta \to \infty$ at one fixed small $n$.

An "original reserve" means the actual energy $\beta \sum_{p \in Y} s_p$ already retained in the Wilson measure, for the indicated completed microscopic set $Y$. It is not a fresh independent energy budget. A field-independent saved source factor multiplies every member of its source packet. A field-dependent coefficient is differentiated and is outside the bare-source theorem unless its derivatives are separately controlled.

## 2 The local moment input and its precise finite-size scope

The first improvement comes from combining two parts of the source estimates [W] whose scopes overlap: the local plaquette estimate of Chapter 10 and the physical contact calculation of Chapter 155.

Let $m = \dim SU(N) = N^2 - 1$. The spanning-tree lower bound and temporal-chain upper bound of Chapter 10 have the form

$$(c_- \beta^{-m/2})^{Q_n} \leq Z_\beta \leq (c_+ \beta^{-m/2})^{R_n}, \quad Q_n = 3n^4 + 1, \quad R_n = 3n^4 - 4n^3 + 1. \tag{3}$$

The fixed action normalization changes $c_\pm$ only. The lower bound follows by fixing a spanning tree and restricting the remaining $Q_n$ link variables to balls of radius $O(\beta^{-1/2})$. The upper bound discards nonnegative plaquette actions and integrates chains, each supplying a one-link integral bounded by $c_+ \beta^{-m/2}$. The difference $Q_n - R_n = 4n^3$ must be retained.

Choose a fixed compatible chessboard cell of volume $v$ with an observed plaquette strictly inside. The number of cells is $M_n = n^4/v$. Wilson reflection positivity and its chessboard consequence [5, 6] give,

for $0 \le t < 1$,

$$\mathbb{E}_\beta e^{t\beta s_p} \le \left( \frac{Z_{(1-t)\beta}}{Z_\beta} \right)^{1/M_n} \le c_+^{R_n/M_n} c_-^{-Q_n/M_n} (1-t)^{-mR_n/(2M_n)} \beta^{2mv/n}. \tag{4}$$

For the first inequality the disseminated plaquettes form a subset of the full plaquette set, so their action is bounded by $S_0$. Translations of the board cover every original orientation and pointing. Equation (2) bounds the last factor without deleting it.

**Lemma 2.1** (Uniform local energy moments)**.** *For a fixed $t_0 \in (0,1)$, the hypotheses above give a constant $A = A(N, v, K, t_0)$ such that*

$$\sup_p \mathbb{E}_\beta e^{t_0 \beta s_p} \le A, \qquad \|\beta s_p\|_{L^u(\mu_\beta)} \le Cu \quad (u \ge 1). \tag{5}$$

*Consequently, for every original finite plaquette set $Y$,*

$$\left\| \beta \sum_{p \in Y} s_p \right\|_{L^u(\mu_\beta)} \le Cu\, \#Y. \tag{6}$$

*Proof.* The exponential bound is (4) at $t_0$. For an integer $u \ge 1$, the inequality $x^u \le u! t_0^{-u} e^{t_0 x}$ gives the moment bound. For other $u$, use the next integer and monotonicity of $L^u$ norms. The set bound is Minkowski's inequality. No independence or decay of correlations is assumed. □

The log-free local force estimate used later follows immediately. If $F_i = X_i S_0$ and $F_{i;\mu} = X_i S_0^{(\mu\nu_b)}$ is its original directional part, compactness and the plaquette first-derivative bound give

$$|F_i|^2 + \sum_{\mu \ne \nu_b} |F_{i;\mu}|^2 \le C \sum_{p \ni b} s_p, \qquad \mathbb{E}_\beta |F_{i;\mu}|^2 \le C\beta^{-1}. \tag{7}$$

The last estimate improves the cruder $C \log(2+\beta)/\beta$ input of PD12.31 in this specified regime. It does not estimate a spatial sum of connected covariances.

# 3 Complete bare physical contacts at every fixed order

## 3.1 The physical source is unchanged

For one literal pointed member of the original hypercubic average, define

$$\begin{aligned} P[f] &= \sum_i f_i F_i^2, & D[f] &= \sum_i f_i \sum_{\mu \ne \nu_b} F_{i;\mu}^2, \\ \ell[f] &= \sum_i f_i X_i^2 S_0, & W_f &= \sum_i f_i F_i X_i. \end{aligned} \tag{8}$$

The test weights are field-independent and color-independent. Write $\sigma = \beta a^{-2}/c_\beta$. The exact physical decomposition of PD12.21 is

$$\begin{aligned} O_f &= \sigma\{D[f] - P[f]/3\}, \\ V_f &= -\frac{a^{-2}}{3c_\beta} W_f = -\frac{\sigma}{3\beta} W_f, \\ t_f &= -\frac{\sigma}{3} P[f] + \frac{\sigma}{3\beta} \ell[f], & r_f &= \sigma D[f] - \frac{\sigma}{3\beta} \ell[f], \\ O_f &= t_f + r_f, & D_{V_f} \rho &= \rho t_f, \qquad D_V u = -\operatorname{div}_\lambda(uV). \end{aligned} \tag{9}$$

For the actual symmetrized source, average all four objects $O_f, V_f, t_f, r_f$ with their original pointings, shifted tests, and transport records. Multilinear coefficients then contain the corresponding multiple average. Distinct shifted pointings are not identified before this average.

Let $\eta_1, \dots, \eta_r$ be commuting squarefree variables: $\eta_i^2 = 0$. The original prescription is

$$A = \sum_i \eta_i O_{f_i}, \qquad V = \sum_i \eta_i V_{f_i}, \qquad t = \sum_i \eta_i t_{f_i}. \tag{10}$$

In particular, there is no bare mixed-source compensator. All exponentials of source-dependent differential operators below are finite polynomials.

**Proposition 3.1** (Exact source-preserving conjugation)**.** *Set*

$$B = e^V A - \int_0^1 e^{sV} t \, ds. \tag{11}$$

*Then, as finite source polynomials,*

$$e^{-D_V}(\rho e^A) = \rho e^B, \qquad \int \rho e^A F \, d\lambda = \int \rho e^B e^V F \, d\lambda. \tag{12}$$

*If $I$ is a set of $r \geq 2$ distinct labels, the coefficient $B_I = [\prod_{i \in I} \eta_i] B$ is*

$$\begin{aligned} B_I = \frac{1}{(r-1)!} \sum_{\pi \in S(I)} V_{\pi_1} \cdots V_{\pi_{r-1}} O_{\pi_r} \\ - \frac{1}{r!} \sum_{\pi \in S(I)} V_{\pi_1} \cdots V_{\pi_{r-1}} t_{\pi_r}. \end{aligned} \tag{13}$$

*Every operator acts on all coefficients to its right.*

*Proof.* For a smooth $f$, $D_V(\rho f) = \rho(tf - Vf)$. If $e^{-sD_V}(\rho e^A) = \rho e^{B(s)}$, its differential equation is $\partial_s B(s) = VB(s) - t$, $B(0) = A$. Its finite-polynomial solution is $B(s) = e^{sV} A - \int_0^s e^{uV} t \, du$. This proves the first identity. Since $D_V$ is the density adjoint of $V$, integration against $e^V F$ gives the second. In (11), only $V^{r-1} A/(r-1)!$ and $-V^{r-1} t/r!$ contribute to source degree $r$; expanding the squarefree coefficient gives (13). □

At degree one, $B_i = r_i$. At degree two,

$$B_{12} = \tfrac{1}{2}(V_1 t_2 + V_2 t_1) + V_1 r_2 + V_2 r_1. \tag{14}$$

This is the complete physical coefficient of [W], not its isolated pure-current part. For three differently labeled copies of the same source, (13) gives $B_{123} = 3V_f O_f - V_f t_f$. More generally, $r$ identical copies with distinct labels give $r V_f^{r-1} O_f - V_f^{r-1} t_f$. These checks fix the factorial convention.

### 3.2 The force-factor invariant

For any fixed ordered derivative of a single plaquette action, compactness of $SU(N)$ gives a uniform bound. The derivative vanishes unless all differentiated bond coordinates belong to that plaquette. For a first derivative there is the stronger estimate

$$|X_i s_p| \leq C \sqrt{s_p}. \tag{15}$$

One way to see it is to write $s(U) = \frac{1}{2} \|U - I\|_{\mathrm{HS}}^2$ on $SU(N)$ and differentiate along a bounded invariant tangent generator. Summing over the finitely many plaquettes incident to a bond gives (7).

**Lemma 3.2** (Two first-force factors survive)**.** *For $k \geq 0$, consider the fully expanded expressions*

$$W_{f_1} \cdots W_{f_k} P[h], \qquad W_{f_1} \cdots W_{f_k} D[h].$$

*Every term contains at least two factors that are first derivatives of local Wilson actions. Every other factor is a bounded fixed-order action derivative. For $W_{f_1} \cdots W_{f_k} \ell[h]$ the same assertion holds for $k \geq 1$.*

*Proof.* The seeds $P$ and $D$ each have two first-force factors. Applying $W_f$ adds its coefficient $F_i$ and then differentiates one factor of the old product. It can consume at most one existing first-force factor, so their number stays at least two. Differentiating a higher action derivative consumes none. Repeating the argument proves the assertion without commuting invariant derivatives.

The scalar seed needs its actual Casimir identity. With all original plaquette incidences retained,

$$\ell[h] = C_F \sum_p h_p (N - s_p), \qquad h_p = \sum_{b \in \partial p} h_b. \tag{16}$$

Therefore

$$W_f \ell[h] = -C_F \sum_i \sum_{p \ni b_i} f_i h_p F_i X_i s_p. \tag{17}$$

This first step creates two first-force factors. The preceding induction then applies. The constant in (16) is kept until differentiated; it has not been discarded from the first-order packet. □

The lemma applies at every fixed derivative order. It supplies no uniform estimate for the growth of its constants with $k$.

### 3.3 Connected microscopic supports and the contact count

Use the bounded-degree graph on original bond anchors in which two anchors are adjacent when the relevant local Wilson stencils meet. Include the original finite pointing shifts in these stencils. Equivalently, a fixed finite-range enlargement of this graph may be declared once and then retained throughout the count.

For $I = \{1, \ldots, r\}$, let $\mathcal{C}_I$ be the set of labeled original-coordinate tuples $(i_1, \ldots, i_r)$ whose anchor graph is connected, with the pointings included when the physical average is expanded. Repeated anchors are permitted. For such a tuple let $Y_{\boldsymbol{i}}$ be the union of its complete original plaquette stars and fixed pointing stencils. At fixed $r$, $\#Y_{\boldsymbol{i}} \leq C_r$.

Define the weighted contact count

$$\mathcal{N}_I = \sum_{\boldsymbol{i} \in \mathcal{C}_I} \prod_{j=1}^{r} |f_j(b_{i_j})|. \tag{18}$$

The original colors, orientations and pointings are included in this convention. Each labeled tuple is counted once; permutation coefficients, Leibniz multiplicities and the bounded size of $Y_{\boldsymbol{i}}$ belong to $C_r$. For an average with weights, its actual product weights are included in the summand. One must not count these factors a second time.

**Lemma 3.3** (Locality of the coefficients)**.** *Every nonzero term of (13) is supported on a tuple in $\mathcal{C}_I$. If the source anchor sets split into two groups that cannot be connected by $r - 1$ edges of the declared microscopic graph, then $B_I = 0$.*

*Proof.* The innermost source has one original anchor and its finite local action factors. In a nonzero term, the next invariant derivative must strike a factor depending on its bond coordinate. That coordinate shares a local action stencil with one already present. Its multiplying force adds only its own star. Induction attaches each new labeled anchor to the existing connected set. A disconnected tuple cannot occur. □

For tests supported in a fixed completed microscopic region $Y$, bounded degree gives

$$\mathcal{N}_I \le C_r \, \#Y \prod_{j=1}^{r} \|f_j\|_\infty . \tag{19}$$

For example, sum over a labeled spanning tree and root one label at an anchor in $Y$. At fixed $r$, every other anchor has only a fixed number of choices along each tree edge. This gives a bound linear in the original support count. It is not a product of $r$ separate volumes.

**Theorem 3.4** (One-reserve bound for the complete bare $H$ contacts)**.** *For every fixed $r \ge 2$, the coefficient (13) of the original source satisfies, on real compact Wilson fields,*

$$|B_I(U)| \le C_r a^{-2r} (\beta+1) \sum_{i \in \mathcal{C}_I} \left( \prod_{j=1}^{r} |f_j(b_{i_j})| \right) \sum_{p \in Y_i} s_p(U). \tag{20}$$

*The constant depends on the fixed group, $c_\beta$, order, and declared original stencils. No coarse replacement of $a$ or new reserve is used.*

*Proof.* Each $V_f$ contributes $a^{-2}/(3c_\beta)$. The $P, D$ portions of $O_f, t_f$ contribute $\beta a^{-2}$ times fixed constants, while the $\ell$ portion contributes $a^{-2}$. There are $r-1$ vector fields. Thus the complete coefficient has the prefactor bounded by $C_r a^{-2r}(\beta+1)$.

By Lemma 3.2, every remaining monomial has at least two first-force factors. All other factors, including any additional first-force factors, are bounded on the compact group at fixed order. The two retained factors obey $|uv| \le (|u|^2 + |v|^2)/2 \le C \sum_{p \in Y_i} s_p$ by (15) and finite incidence. Lemma 3.3 gives the connected set of tuples. The finite permutation and product-rule sums are absorbed in $C_r$ as specified in (18). □

**Corollary 3.5** (Log-free absolute moments)**.** *In the torus regime (2), for every fixed $r \ge 2$ and every $u \ge 1$,*

$$\boxed{\|B_I\|_{L^u(\mu_\beta)} \le C_r u \, a^{-2r} \mathcal{N}_I.} \tag{21}$$

*In particular, $\mathbb{E}_\beta |B_I| \le C_r a^{-2r} \mathcal{N}_I$.*

*Proof.* Use (20), Minkowski's inequality, and Lemma 2.1. The factor $(\beta+1)/\beta$ is at most two, and $\#Y_i$ is bounded at fixed $r$. □

The same proof, using Jensen's inequality on the nonnegative weighted plaquette sum in (20), yields constants $c_r, A_r > 0$ such that, if $\mathcal{N}_I > 0$,

$$\mathbb{E}_\beta \exp\left( \frac{c_r |B_I|}{a^{-2r} \mathcal{N}_I} \right) \le A_r. \tag{22}$$

This is an absolute contact estimate, not a concentration estimate with a square-root volume normalization.

For completeness, let $J_r = \sup_{\boldsymbol{i}} \#Y_{\boldsymbol{i}}$. The weights

$$w_{\boldsymbol{i},p} = \frac{\prod_j |f_j(b_{i_j})|}{J_r \mathcal{N}_I} \quad (p \in Y_{\boldsymbol{i}})$$

sum to at most one. Add the remaining weight at zero. The bound (20) and $(\beta+1)/\beta \le 2$ reduce (22), for $c_r \le t_0/(2C_r J_r)$, to Jensen's inequality and (5). Thus no independence or extra reserve is hidden in the exponential estimate.

For fixed physical support of volume $|Y|_a = a^4 \#Y$, (19) gives

$$\|B_I\|_{L^u} \le C_r u\, a^{-(2r+4)} |Y|_a \prod_j \|f_j\|_\infty . \tag{23}$$

The inverse powers remain explicit. This estimate does not say that bare coincident products have a finite continuum limit.

### 3.4 Degree two and original absolute masses

The exact degree-two coefficient is

$$b_{HH}(f,h) = \frac{\sigma^2}{9\beta}\Big\{ \frac{1}{2}(W_f P[h] + W_h P[f]) - 3(W_f D[h] + W_h D[f]) + \frac{1}{2\beta}(W_f \ell[h] + W_h \ell[f]) \Big\}. \tag{24}$$

The ordinary directional-force responses and the positive sign of the last term are essential. If $Y$ is its literal completed overlap, finite incidence and (21) give

$$\boxed{\mathbb{E}_\beta |b_{HH}(f,h)| \le C a^{-4} \#Y \ \|f\|_\infty \|h\|_\infty .} \tag{25}$$

This removes the factor $1 + \sqrt{\log(2+\beta)}$ from PD12.30 within (2). Against the raw measure, the bound is multiplied by the genuine $Z_\beta$. An independent scalar $e^s$ contributes $e^{\Re s}$. If a label carries a field-independent saved factor $\Gamma_i$, the entire coefficient and bound carry $\prod_i \Gamma_i$ and $\prod_i |\Gamma_i|$, respectively, exactly once.

The theorem applies at birth on the original common mesh. A later effective coefficient may have field-dependent prefactors, old meshes, other density scores, and moving constraints. Those terms are not covered by merely replacing $a$ with the current coarse scale.

## 4 What positive transport preserves, and what it does not

**Proposition 4.1** (Absolute contact mass under complete positive transport)**.** *Let $T$ be a source-free positive mass-preserving kernel from the original complete physical configuration space to an output space. Include every selector alternative and all factors needed to make it the actual measure transport. For the single signed input measure $\nu_I = e^s B_I \rho\, d\lambda$, with $s$ an independent field-free scalar,*

$$\|T_* \nu_I\|_{\mathrm{TV}} \le e^{\Re s} Z_\beta \mathbb{E}_\beta |B_I| \le e^{\Re s} Z_\beta C_r a^{-2r} \mathcal{N}_I. \tag{26}$$

*The same statement holds for finite compositions of such complete kernels.*

*Proof.* For $|F| \le 1$, positivity and $T1 = 1$ imply $|TF| \le 1$. Hence $|\nu_I(TF)| \le \int |\nu_I|$. Taking the supremum over $F$ proves total-variation contraction. The last estimate is Corollary 3.5. □

There are two distinct qualifications. First, the coefficient of the *logarithm* of an integrated source functional is not generally $T_*(\rho B_I)/(T_*\rho)$: it contains conditional cumulants and products of lower source coefficients. Second, source conjugation changes the test carried through the kernel. Proposition 3.1 gives the tested identity

$$\int \rho e^A \, K\Psi \, d\lambda = \int \rho e^B e^V (K\Psi) \, d\lambda. \tag{27}$$

It does not permit replacement of $T_*(\rho e^A)$ by $T_*(\rho e^B)$ while holding the retained output fixed. The derivatives of $K\Psi$ include the original numerator, denominator, selectors, boundary currents, coarea factors, live references and effective score.

Thus (26) is useful for an actually transported bare signed density, with its original mass and mesh. It is not a bound on a signed localized polymer norm, and it does not close all-history MT36. A selected branch lacking unit mass cannot use its own normalization without retaining that normalization explicitly.

# 5 The remaining physical susceptibility and an averaged bound

## 5.1 An exact fixed-source target

For the uniform real test, retain PD12.39:

$$G_H = \sum_{b,A} \left\{ \sum_{\mu \neq \nu_b} (F^A_{b;\mu})^2 - \frac{1}{3}(F^A_b)^2 \right\} \ge 0, \qquad O_H = \frac{\beta a^{-2}}{c_\beta} G_H. \tag{28}$$

The sum is the directional variance $\sum_\mu |F_{b;\mu} - F_b/3|^2$. Finite incidence gives

$$0 \le G_H \le C_H S_0. \tag{29}$$

Define the actual nonlinear source family

$$Z_H(\beta, \lambda) = \int e^{-\beta S_0 + \lambda \beta G_H} \, d\lambda_{\mathrm{Haar}}, \qquad q_{n,\beta}(\lambda) = \frac{1}{n^4} \log Z_H(\beta, \lambda). \tag{30}$$

The measure in the integral is Haar measure; the scalar $\lambda$ is the source parameter. Equivalently it is the original $O_H$ source with parameter $\eta = \lambda c_\beta a^2$. Let $\mu_{\beta,\lambda}$ denote its normalized positive real measure. At finite cutoff,

$$q'_{n,\beta}(\lambda) = \frac{\beta}{n^4} \mathbb{E}_{\beta,\lambda} G_H \ge 0, \qquad q''_{n,\beta}(\lambda) = \frac{\beta^2}{n^4} \operatorname{Var}_{\mu_{\beta,\lambda}}(G_H) \ge 0. \tag{31}$$

Therefore the desired matching extensive bare upper bound is exactly

$$\boxed{\sup_{n,\beta} q''_{n,\beta}(0) < \infty} \tag{32}$$

in the stated continuum regime. Indeed (32) gives $\operatorname{Var}(O_H) \le C c_\beta^{-2} a^{-4} n^4 = C T^4 a^{-8}$, matching the order of PD12.40's lower bound. This one scalar bound would still be weaker than the full graded marked induction.

The contact estimate has not proved (32). The exact identity remains

$$\operatorname{Var}(O_H) = \operatorname{Var}(r_H) + \mathbb{E} b_{HH}. \tag{33}$$

Equation (25) controls the second term. The first is an ordinary connected covariance. The total-force Ward identity does not turn the directional-force expression $D$ into a complete current square.

### 5.2 A bound that does hold for the actual tilted family

**Theorem 5.1** (Uniform source-averaged $H$ susceptibility)**.** *Under (2), let $\lambda_* = 1/(2C_H)$. There is a constant $B$ independent of cutoff and torus size such that*

$$\boxed{\int_{-\lambda_*/2}^{\lambda_*/2} \frac{\beta^2}{n^4} \operatorname{Var}_{\mu_{\beta,\lambda}}(G_H)\, d\lambda \le \frac{2B}{\lambda_*}.} \tag{34}$$

*This is a statement about the original Wilson source functional with a real $H$ source. It is not a claim about its susceptibility at zero.*

*Proof.* From (29), $Z_H(\beta, \lambda_*) \le Z_{\beta/2}$. Positivity of $G_H$ implies

$$0 \le q(\lambda_*) - q(0) \le \frac{1}{n^4} \log \frac{Z_{\beta/2}}{Z_\beta}. \tag{35}$$

The matching powers (3) give

$$\frac{1}{n^4} \log \frac{Z_{\beta/2}}{Z_\beta} \le \frac{R_n}{n^4} \log c_+ - \frac{Q_n}{n^4} \log c_- + \frac{mR_n}{2n^4} \log 2 + \frac{2m \log \beta}{n} \le B. \tag{36}$$

All exponents and the finite-size term have been kept. Constants may be enlarged to make $B > 0$.

The real function $q$ is increasing and convex. Thus

$$q'(\lambda_*/2) \le \frac{q(\lambda_*) - q(\lambda_*/2)}{\lambda_*/2} \le \frac{2B}{\lambda_*}, \qquad q'(-\lambda_*/2) \ge 0.$$

Integrating $q''$ over the interval and using (31) proves (34). □

For any $A > 0$, the set of source parameters in this interval with $\beta^2 \operatorname{Var}(G_H)/n^4 > A$ has length at most $2B/(\lambda_* A)$. The bad interval may nevertheless contain zero for every cutoff. Choosing a different source value changes the measure and cannot be substituted into the original construction without a further argument.

### 5.3 Why real bounds do not close the fixed-source gap

A simple diagnostic separates these assertions. On $M$ sites let a single Bernoulli variable $X \in \{0, 1\}$ have equal probabilities, and set $y_x = X$ at every site. The law is translation invariant and reflection positive: for any positive-side function $F$, its reflected quadratic form is $\mathbb{E}|F(X)|^2 \ge 0$. All local exponential moments are uniformly bounded. Nevertheless,

$$\operatorname{Var}\left(\sum_x y_x\right) = M^2/4, \qquad q_M(\lambda) = \frac{1}{M} \log \frac{1 + e^{M\lambda}}{2}, \qquad q_M''(0) = M/4. \tag{37}$$

The real pressures are uniformly Lipschitz on fixed intervals, while the nearest complex zeros approach zero at distance $\pi/M$. This is not a counterexample to Wilson Yang–Mills. It proves that local moments, reflection positivity, and real pressure stability alone cannot imply (32).

A verified zero-free complex source disc with a uniformly bounded pressure logarithm would imply a fixed-source derivative estimate by Cauchy. Finite-cutoff analyticity does not supply such a disc. This remains a model-specific analytic task.

# 6 Native scale crossing and the slower summation problem

NM12 of [W] records a genuine mismatch: the summed majorant in the 1985 variational paper has rate $\delta_0/2$, while its nonlinear derivative estimate (190) has rate $\delta_0/8$. Weakening the latter rate does not create a summation estimate. The argument here instead derives a bound from the nested geometry.

The accessible metric formulation is Dimock [7], printed p. 13, (88)–(92), which states the collar separation and shell-weighted path length and attributes its summability to Bałaban's 1984 Lemma 2.1 [8]. The following proof is geometric and works in dimension four; no three-dimensional field estimate is imported.

## 6.1 A theorem requiring only charged nested collars

Fix $L > 1$, scales $\sigma_j = L^{j-k}$, and nested spatial domains $\Omega_0 \supset \cdots \supset \Omega_k$. Put $A_j = \Omega_j \setminus \Omega_{j+1}$ for $j < k$ and $A_k = \Omega_k$. Assume

$$\operatorname{dist}(\Omega_{j+1}, \Omega_j^c) \geq \rho\sigma_j \tag{38}$$

for one fixed $\rho > 0$. Let the cost of every admissible spatial path include

$$\sum_{j=0}^{k} \sigma_j^{-1} \operatorname{length}(\gamma \cap A_j). \tag{39}$$

Other path restrictions or nonnegative additional costs are allowed. Let $d_\Omega$ be the resulting infimum over paths.

**Lemma 6.1** (Depth costs native distance)**.** *If $x \in \Omega_k$ and $y \in A_j$, then*

$$d_\Omega(x, y) \geq \rho \max(k - j - 1, 0). \tag{40}$$

*This assertion does not require a positive cost outside the outermost domain.*

*Proof.* Only $k - j \geq 2$ needs proof. A path from $x$ to $y$ must leave $\Omega_\ell$ for every $\ell = j + 1, \ldots, k - 1$. Before its first exit from $\Omega_\ell$, take its last visit to the boundary of $\Omega_{\ell+1}$ coming from that inner set. The intervening subpath lies in $A_\ell$ and has length at least $\rho\sigma_\ell$ by (38). Its weighted cost is at least $\rho$. These portions lie in distinct shells and in disjoint chronological intervals. There are $k - j - 1$ of them. Infimizing proves the claim. Discrete endpoints can equivalently be treated by the same closed-cell path convention as the separation condition; any lattice endpoint loss must be included in the chosen $\rho$. □

**Corollary 6.2** (Scale compensation)**.** *For $s \geq 0$, a top-scale $x$, and $y \in A_j$,*

$$\boxed{\sigma_y^{-s} e^{-c d_\Omega(x,y)} \leq L^s e^{-(c - s \log L/\rho) d_\Omega(x,y)}.} \tag{41}$$

*Thus the assigned-scale factor is uniformly controlled if $c\rho \geq s \log L$; a strict inequality leaves an exponential margin.*

*Proof.* Writing $m = k - j$, Lemma 6.1 gives $m \leq 1 + d_\Omega/\rho$. Hence $\sigma_y^{-s} = L^{sm} \leq L^s e^{s(\log L) d_\Omega/\rho}$. □

### 6.2 A slower sum retaining the actual determining volume

Let $\mathcal{B}_j$ be any subset of the original grid of spacing $\sigma_j$ in $A_j$. Suppose the complete determining set is covered by $N_D$ actual unit cubes in current-scale units. This is a physical covering count, not the number of microscopic links.

**Theorem 6.3** (Finite-domain native sum)**.** *For a top-scale $x$, $q \geq 0$, and $\gamma\rho > (d+q)\log L$,*

$$\sum_{j=0}^{k} \sum_{y \in \mathcal{B}_j} \sigma_y^{-q} e^{-\gamma d_\Omega(x,y)} \leq C_d N_D \left[1 + \frac{L^{d+q}}{1 - L^{d+q} e^{-\gamma\rho}}\right]. \tag{42}$$

*The actual orientation or component multiplicity multiplies the right side. For $\gamma > \gamma_0 > (d+q)\log L/\rho$, restriction to $d_\Omega(x,y) \geq D$ gives the right side at $\gamma_0$ multiplied by $e^{-(\gamma-\gamma_0)D}$.*

*Proof.* At depth $m = k - j$, each unit cube contains at most $C_d L^{dm}$ grid points, and $\sigma_y^{-q} = L^{qm}$. The depth-zero term is at most $C_d N_D$. For $m \geq 1$, (40) bounds the contribution by $C_d N_D L^{(d+q)m} e^{-\gamma\rho(m-1)}$. Sum this geometric series. The tail follows by splitting $e^{-\gamma d}$ into factors with rates $\gamma_0$ and $\gamma - \gamma_0$. □

For $d = 4$, $c = \delta_0/8$, and $\gamma_0 = \delta_0/16$, an unweighted tail is available if

$$\rho > \frac{64 \log L}{\delta_0}. \tag{43}$$

The explicit $N_D$ remains. This is a sufficient threshold, not a claim of optimality.

### 6.3 What a volume-uniform sum additionally requires

Suppose, in addition, that every physical path segment is charged by (39), or that admissible paths stay in the charged shell union. Since all $\sigma_j \leq 1$, this implies

$$\mathrm{dist}_{\mathrm{spatial}}(x,y) \leq d_\Omega(x,y). \tag{44}$$

The condition is separate from Lemma 6.1.

**Theorem 6.4** (Volume-uniform native sum under complete path cost)**.** *With (44), for every $q, s \geq 0$ and $\gamma > (d+q)\log L/\rho$,*

$$\sum_{j} \sum_{y \in \mathcal{B}_j} \sigma_y^{-q} (1 + d_\Omega(x,y))^s e^{-\gamma d_\Omega(x,y)} \leq C_{d,L,\rho,q,s,\gamma}, \tag{45}$$

*uniformly in cutoff, history, and torus volume, for a top-scale anchor.*

*Proof.* A metric ball of radius $t$ lies in a spatial ball of radius $t$ by (44). At depth $m$, it contains at most $C_d(1+t)^d L^{dm}$ grid points. Also $m \leq 1 + t/\rho$. Thus its weighted count satisfies

$$W_q(t) \leq C_{d,L,q}(1+t)^d \exp\left(\frac{(d+q)\log L}{\rho} t\right). \tag{46}$$

Split the sum into shells $h \leq d_\Omega < h+1$ and sum the resulting polynomial times $\exp[-(\gamma - (d+q)\log L/\rho)h]$. Periodic representatives give the same spatial count on a torus. □

The accessible wording of Dimock (91) lists paths in the torus but writes the cost only over the displayed shells. It does not explicitly specify the exterior convention on that page. A literal zero-cost exterior would invalidate (44) and global summability: take many separated components, travel out of one and into another at bounded charged cost. The intended convention may exclude such paths or charge the exterior, but this note does not insert it silently. The original 1984 definitions (2.47)–(2.51), their link to the 1985 metric, and the status of the outermost region still need direct primary-text verification. Theorems 6.3 and 6.4 are therefore kept separate.

## 7 Native response across lower shells

This section states a consequence for an *identified* native reference family, using the response structure recorded in [W], PE12, NB12, UW12 and NM12. It does not identify the three-join mixed numerator with that family.

Let

$$U_C = e^{iaH(C)}U_0, \qquad H(0) = 0, \qquad C \in \mathbb{C}^{n_{\mathrm{ref}}}, \qquad n_{\mathrm{ref}} = (\dim G)q_\Lambda, \tag{47}$$

where the input coordinates are the original top-scale free target coordinates. For every target used below, require that the full segment $tC$, $0 \le t \le 1$, stays in one admitted branch; an admitted centered target ball suffices. Suppose uniformly on that branch the native response is

$$\begin{aligned} |\partial_{C_i} H(C)(b)| &\le K_{\mathrm{nat}} \sigma_y^{-1} e^{-c d_{\mathrm{nat}}(x_i, y)}, \\ |D_{a,U_0} \partial_{C_i} H(C)(b)| &\le K_{\mathrm{nat}} \sigma_y^{-2} e^{-c d_{\mathrm{nat}}(x_i, y)}. \end{aligned} \tag{48}$$

The output bond and its complete derivative stencil lie in the actual assigned cell at $y$. These are the potential and first-difference weights of the original (190) with top-scale input, not a response asserted for arbitrary old physical currents.

To apply Section 6 one must verify that this original distance satisfies its collar-cost lower bound. If only $d_{\mathrm{nat}} \ge \kappa d_\Omega$ is known, replace $\rho$ by $\rho_{\mathrm{eff}} = \kappa\rho$ throughout. Neither $\kappa$ nor the dependence of $\delta_0$ on the fixed parameters is guessed here.

**Theorem 7.1** (Finite-record response with lower-shell outputs)**.** *Assume the original response distance has effective collar constant $\rho_{\mathrm{eff}}$ and*

$$c\rho_{\mathrm{eff}} > 2\log L, \qquad \alpha = c - \frac{2\log L}{\rho_{\mathrm{eff}}} > 0. \tag{49}$$

*Then every actual input direction $h$ satisfies, on a complete output record $D^+$,*

$$\|D_C H(C)[h]\|_{W^{1,\infty}_{a,U_0}(D^+)} \le K_{\mathrm{nat}}(L + L^2)\,\|h\|_{\ell^1}\,, \tag{50}$$

$$\|D_C H(C)[h]\|_{W^{1,p}_{a,U_0}(D^+)} \le C_p \nu^{1/p} v_a(D^+)^{1/p}\,\|h\|_{\ell^1}\,. \tag{51}$$

*Here $v_a(D^+) = a^4 N_{D^+}$ is the original fine volume of the complete record and $\nu$ is any actual bounded-overlap multiplicity used in the assignment. On the original cutoff $C = gB$, $\|B\|_\infty \le R = M_0 p_0(g)$,*

$$\boxed{\|H(gB)\|_{W^{1,p}_{a,U_0}(D^+)} \le C_p \nu^{1/p} v_a(D^+)^{1/p} n_{\mathrm{ref}}\, g M_0 p_0(g).} \tag{52}$$

*If the complete output record is at native distance at least $D$ from every input anchor, the bounds gain $e^{-\alpha D}$.*

*Proof.* Apply (41) to the two factors in (48), with $s = 1, 2$. Both are bounded by their displayed $L$ factors times $e^{-\alpha d_{\mathrm{nat}}}$. Contract with the actual $h_i$ and sum absolute values. This proves (50). Summing the $p$th powers with the original $a^4$ volumes proves (51). Integrate the derivative along $tC$, $0 \le t \le 1$, then use $\|C\|_1 \le n_{\mathrm{ref}} g R$. The distance margin proves the final assertion. □

No global metric-volume estimate was used. For $c = \delta_0/8$, the sufficient threshold is $\rho_{\mathrm{eff}} > 16 \log L/\delta_0$. This is weaker than (43). These must be inequalities for the actual originally fixed constants. They cannot be justified by enlarging a history buffer after that history has been selected.

If the map is holomorphic on a fixed target sup-norm ball of radius $r_*$ and both the potential and covariant first-difference bounds in (48) are uniform there, Cauchy estimates on a smaller ball give, for fixed $r \ge 1$,

$$\|D_C^r H(C)[h_1, \dots, h_r]\|_{W^{1,p}_{a,U_0}(D^+)} \le C_{p,r} r_*^{1-r} \nu^{1/p} v_a(D^+)^{1/p} \times \|h_1\|_1 \prod_{j=2}^{r} \|h_j\|_\infty . \tag{53}$$

The first direction may be chosen by symmetry. Passing from $C$ to $B$ contributes $g^r$. An actual coordinate-tensor contraction retains its full coordinate sum; replacing it by a largest-entry norm incurs the appropriate power of $n_{\mathrm{ref}}$.

### 7.1 The strict-margin weak-chart implication

Suppose $U_0 = e^{iaA_0}$ is already represented in the *same fixed anchored chart* on $D^+$ with $p > 8$ and $\|A_0\|_{W^{1,p}_a} \le \tau_0 < \tau$. Assume this established chart has its stated uniform product and Sobolev bounds. On its common small branch, the product/logarithm map and the lattice product rule give

$$\|A(C) - A_0\|_{W^{1,p}_a(D^+)} \le C_{\tau_0} \|H(C)\|_{W^{1,p}_{a,U_0}(D^+)}, \qquad U_C = e^{iaA(C)}. \tag{54}$$

The covariant-to-ordinary derivative difference is bounded by $C|A_0||H|$, and the product/logarithm series has bounded coefficients on the common branch; no inverse fine-mesh factor is introduced. Therefore the original chart margin is preserved if

$$C_{\tau_0} C_p \nu^{1/p} v_a(D^+)^{1/p} \|C\|_1 < \tau - \tau_0. \tag{55}$$

This is a perturbative admission statement for the identified family. It does not construct the baseline anchored chart or match independent numerator walls to the coordinates $C$.

## 8 Limits of protection by unchanged target rows

The exact protected sector of [W] remains useful: a coefficient whose entire saved dependence factors through unchanged complete target rows is constant on the corresponding auxiliary fiber. The complete common-row identity is established for the actual numerator in R-II (1.50) and the final new minimum. Equations (1.51)–(1.58) then make successive background replacements [10]; that fact does not establish the same identity for every intermediate join or localized field. The property must be checked for each full source dependence record.

There is a concrete reason it cannot be extended to all normalized physical marks merely because the raw field perturbation is small. On a flat admitted native reference with genuine interior free

coordinates, the positive axial quadratic form gives the local action expansion

$$O_g(B) := g^{-2}\{A(U_{gB}) - A(U_0)\} = \tfrac{1}{2}\langle B, Q_{\mathrm{nat}} B\rangle + O(g|B|^3), \qquad Q_{\mathrm{nat}} \geq \lambda_M I. \tag{56}$$

This is the native flat-reference expansion behind R-II (1.2)–(1.9), in its admitted finite-dimensional scope. Take one fixed nonzero $v$ within this target chart and the original cutoff. For sufficiently small $g$,

$$|O_g(v) - O_g(0)| \geq \lambda_M |v|^2/4. \tag{57}$$

All complete common exterior rows have the same values at $B = 0$ and $B = v$. Consequently any function $F$ of those rows obeys

$$\max_{B\in\{0,v\}} |O_g(B) - F| \geq \lambda_M |v|^2/8. \tag{58}$$

The proof is the triangle inequality applied to the two values. An $O(g)$ perturbation of a native field can therefore create order-one dependence in a physically normalized action observable. Common-row measurability and small field variation are different assertions.

This example concerns the normalized action and its positive native Hessian. The $H$ observable requires its own tangent calculation; no positivity conclusion for its native Hessian is inferred. The result is also not a counterexample to AE8/LM47. It limits an attempted shortcut that would identify every physical mark with the protected row algebra.